

BEATRICE, NE, USA

WMO: 725515

Lat: 40.301N

Lon: 96.754W

Elev: 1324

StdP: 14.01

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	3.3	-9.3	3.5	0.3	-4.4	4.6	6.2	33.0	28.2	28.7	24.7	7.4	320	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	96.8	74.4	93.1	73.9	90.5	73.4	78.2	90.2	76.8	88.5	75.4	86.4	12.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	137.0	84.8	73.1	129.3	83.1	72.2	125.0	82.4	42.6	90.3	41.1	88.6	39.8	86.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	24.4	20.9	DB	-10.0	101.4	4.7	2.8	-13.4	103.4	-16.1	105.0	-18.8	106.5	-22.2	108.6	(4)
(5)			WB	-10.6	80.8	4.5	1.6	-13.8	82.0	-16.5	82.9	-19.0	83.8	-22.3	84.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	26.3	30.5	41.5	52.3	63.2	73.4	77.9	75.4	67.4	54.5	40.9	29.4		
	DBStd	20.43	11.90	12.12	11.78	9.76	8.69	6.74	6.35	5.98	8.53	9.62	10.19	10.83		
	HDD50	2703	736	552	314	87	6	0	0	0	2	63	303	640		
	HDD65	5691	1201	966	731	392	140	11	2	2	73	347	723	1103		
	CDD50	3738	0	5	50	156	416	701	866	786	525	201	31	1		
	CDD65	1254	0	0	2	11	85	262	404	324	146	20	0	0		
	CDH74	13303	0	0	32	176	903	2731	4533	3204	1495	221	8	0		
	CDH80	5659	0	0	5	41	304	1118	2172	1394	570	54	1	0		
	WSAvg	9.4	9.4	9.9	10.8	11.7	10.2	9.3	7.7	7.1	8.3	9.3	9.5	9.2		
	PrecAvg	30.4	0.7	0.8	2.1	2.7	4.5	4.2	4.0	3.7	3.3	2.2	1.4	0.8		
(15) Precipitation	PrecMax	44.7	2.1	3.1	7.2	6.9	11.2	8.7	11.6	10.3	11.9	6.3	3.7	2.4		
	PrecMin	13.4	0.0	0.0	0.0	0.1	1.5	0.3	0.1	0.4	0.3	0.0	0.0	0.0		
	PrecStd	6.4	0.5	0.7	1.7	1.4	2.3	1.9	2.4	2.3	2.4	1.6	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	68.9	77.6	85.0	92.8	96.9	101.5	99.6	95.1	86.4	74.2	61.8		
		MCWB	45.4	51.6	59.5	63.1	69.7	73.7	74.3	73.6	72.3	66.9	57.0	51.4		
	2%	DB	53.2	60.7	72.3	79.2	87.6	92.9	97.8	94.9	90.6	80.0	67.9	54.8		
		MCWB	43.2	48.3	56.6	60.4	68.2	73.7	75.3	74.3	71.7	63.0	53.7	45.9		
	5%	DB	46.9	54.0	66.4	73.7	82.6	90.1	93.9	91.1	87.1	75.1	62.9	50.1		
		MCWB	39.8	45.1	54.2	58.4	66.0	73.4	75.1	74.2	70.1	61.2	52.2	42.8		
	10%	DB	42.7	47.7	61.0	69.9	79.1	86.6	90.6	88.1	82.2	71.6	57.4	45.2		
		MCWB	37.2	40.7	51.1	56.7	64.8	71.4	74.4	73.2	67.9	59.1	48.4	39.4		
	0.4%	WB	47.5	56.3	63.7	66.5	73.2	79.0	80.5	79.4	76.6	70.7	60.1	54.8		
		MCDB	57.3	63.5	72.8	78.3	86.4	90.4	92.2	89.5	88.4	80.1	67.1	59.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.0	49.5	60.0	63.5	71.1	76.8	78.5	77.4	74.5	67.1	56.8	47.4		
		MCDB	51.4	59.3	66.8	74.3	82.6	88.3	91.2	88.8	85.1	75.6	65.4	52.5		
	5%	WB	40.7	44.8	55.9	60.7	69.0	75.2	77.3	75.8	72.6	63.8	53.1	42.8		
		MCDB	47.2	52.8	64.8	71.1	79.5	86.4	89.6	86.8	82.4	71.4	61.1	49.5		
	10%	WB	37.2	41.0	51.2	57.8	66.6	73.3	75.8	74.4	70.6	60.3	49.4	39.3		
		MCDB	42.3	47.8	61.5	68.1	76.4	83.9	87.0	84.9	79.9	68.7	57.6	44.8		
	5% DB	MDBR	19.2	19.7	22.2	22.4	21.1	20.3	20.5	20.6	22.9	23.0	21.6	18.9		
		MCDBR	27.6	29.8	30.8	28.5	26.2	23.2	24.0	24.6	25.9	28.0	28.8	27.2		
		MCWBR	18.9	18.9	17.3	15.4	11.8	9.7	8.5	9.7	10.5	14.3	17.2	18.9		
		MCWBR	26.3	27.0	24.9	25.3	22.0	20.8	20.7	21.2	21.5	22.3	25.9	24.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	18.7	18.2	15.5	15.4	11.4	10.5	9.4	9.8	10.5	13.5	17.3	18.1		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.297	0.323	0.372	0.404	0.414	0.421	0.423	0.380	0.334	0.295	0.281		
	taud		2.479	2.450	2.420	2.305	2.284	2.312	2.291	2.292	2.360	2.476	2.511	2.484		
	Ebn at Noon		283	293	296	285	278	274	271	267	272	276	275	274		
	Edh at Noon		24	29	33	40	42	41	41	40	35	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		693	965	1270	1602	1871	2104	2086	1835	1505	1083	770	588		
	RadStd		57	64	104	102	140	134	132	89	99	115	56	56		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page